04/21

Monday, 21 April 2025 9:34 AM

Exam 2 → May 5th 1-3pm [open book, open notes] 12 questions, 5 points
One problem all throughout

HW7 → due May 27th

Upwind difference → need to know HW7

→ Explicit methods biggest problem

time step for stability is very small (stability criteria very stringent & it gets worse)

→ Implicit methods biggest issue

↳ need to have non-linearization
[time linearization → newton-rhapson]

↳ Need a bigger stencil (higher matrix bandwidth) → Need LU methods or ADI (factorization)

↳ How to address time steps

Time-split method

↳ Unconditionally stable for linear [but conditionally stable for non-linear]

Accelerating Convergence Rate

Explicit methods: (How to increase computational speed)
+ residual smoothing
+ local time stepping
+ Multigrid

error is removed quickly (neighbors seen right away) } h then 2h & then 4h

Compressible flow of a non-reacting ideal gas

BC's: Inflow / outflow → use euler equation BC's (momentum dominates over viscous, reynolds no. is high, diffusion in streamwise direction is negligible)

↓

else have to use full Navier stokes

exceptions: high Re, but if grid gets really small then need to have full info